

Exploring the variations in disyllabic lexical tone sandhi in Xiangshan Chinese

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Abstract

Lexical tone sandhi in Northern Wu Chinese languages typically adopts a left-dominant mechanism, usually characterized as a tone extension pattern from the leftmost tone rightwards spanning the whole sandhi domain. By default, the underlying tone of the leftmost tone should be the only major factor conditioning the sandhi outputs in a lexical sandhi domain. However, variations still exist in some languages, the causes of which have not been studied extensively.

This study examines the categorical variations in disyllabic lexical tone sandhi in Xiangshan Chinese, an under-studied Northern Wu variety. A perceptual analysis of 106 disyllabic lexical compounds produced by 8 native Xiangshan speakers revealed 4 distinctive sandhi patterns. These patterns were cross-checked by independent *k*-means clustering based on acoustic data. Further analyses of the cluster distributions suggest that the historical tonal categories of the initial tones, and individual speakers, serve as the major sources of the variations. The results provide evidence supporting the reality and relevance of historical tonal categories in tone sandhi, and suggest distinct development paths for sandhi tones and citation tones. Inter-speaker variations largely explain the remaining variation and may reflect different stages of ongoing language change.

Index Terms: tone sandhi, Northern Wu Chinese, cluster analysis, variations

1. Introduction

Tone sandhi, the phonological changes of tones in conditioned contexts, generally exhibits a *left-dominant* pattern in lexical words in Northern Wu Chinese [1], [2], [3], [4]. It is typically realized as a rightward tone extension of the initial tone over the whole sandhi domain, optionally followed by a default L tone insertion, e.g., in Shanghai [5], [6], [7]. In some varieties, the underlying tone of the initial syllable is replaced with another tone before its features spread rightwards, e.g., in Wuxi [8].

By default, the underlying tone of the leftmost tone should be the only major factor conditioning the sandhi outputs in a lexical sandhi domain. However, distinct patterns persist in some Northern Wu varieties even when the leftmost tone remains constant. For example, in Zhenhai, an LM-initial disyllabic tone sequence in a lexical domain can either surface as L-MH or LM-L. It is claimed that the former has a final stress which attracts the underlying contour tone of the initial syllable; on the other hand, the latter has an initial stress which preserves the initial tone in its original position [9]. However, there is no consistent acoustic cue signalling this “stress” distinction, and it has been further suggested that the co-existence of such two different sandhi realizations might have correlations with historical tonal categories [10]. In ancient Chinese, there were 4 distinctive tonal categories, i.e., *ping* (level), *shang* (rising),

qu (departing), *ru* (checked), each of which further split into *yin* (High) and *yang* (Low) registers in Middle Chinese. Despite disparate tone mergers and changes, the register difference is largely preserved in most Wu dialects. The relevance of historical categories in tone sandhi processes can be found in some Wu dialects, such as Danyang, whose tone sandhi outputs are predicted by the historical rather than synchronic tonal categories of the initial tones [11], [12].

Exceptional cases, where *non*-initial tones play a role in determining the lexical sandhi outputs, also exist in Wu dialects. For example, in Chongming, disyllabic words with the same initial underlying tones show different tone sandhi types depending on whether the second tone is oblique or even [13]. Furthermore, recent experimental studies on tone sandhi in different phrasal structures in Shanghai and Wuxi confirmed that other factors, such as lexical frequency and semantic transparency, can also influence the tone sandhi application [14]. Given the complexities and inconsistencies of the lexical tone sandhi mechanisms across Northern Wu dialects, further investigation is required to understand the potential causes of various distinctive lexical sandhi patterns.

Xiangshan Chinese, an under-studied Northern Wu dialect, has 6 distinctive citation tones, 4 of which are non-checked tones (i.e., HH, HL, LHL, LH) and the remaining 2 are checked tones (i.e., Hq, LHq) [15]. As shown in Table 1, the historical *ping* and *shang* categories have merged into one synchronic tone category in each of the *yin* and *yang* registers. Apart from that, occasional tonal switch was found between the HH and HL tones, as well as between the LHL and LH tones (marked via dashed lines) [16]. For example, the majority of syllables with historical *yinping* and *yinshang* tones is now pronounced with an HH tone, but sometimes an HL tone is produced instead for certain syllables under these categories by some speakers.

Table 1: *Synchronic and historical citation tone system in Xiangshan Wu Chinese.*

	<i>ping</i>	<i>shang</i>	<i>qu</i>	<i>ru</i>
<i>yin</i>	HH		HL	Hq
<i>yang</i>	LHL		LH	LHq

In [17]’s massive data collection across 48 sites in the Wu region, speech recordings include those from 5 different places in Xiangshan in 1970s and 1990s. According to their analysis, the disyllabic lexical words with *yinping-yinping* tone combination in 4 regions in Xiangshan showed 2 distinctive tone sandhi contours, which come from several repetitions of the same words produced by the same speakers. The causes of such variations, however, were not further analysed. In this study, we extend this research line by asking 2 questions: (1) What are the possible distinct variations, if any, in the lexical tone sandhi domain in Xiangshan Wu Chinese? (2) What are the possible factors contributing to such variations?

2. Methods

2.1. Data elicitation

The current analysis examined a subset of data from a larger corpus of Xiangshan Chinese collected by the first author in March and April 2021. In total, the corpus consists of 416 disyllables and 352 trisyllables with varying citation tone combinations and morphosyntactic structures, the selection of which was determined in a pilot study with a male 53-year-old Xiangshan native speaker in 2020 and 2021.

The present data were elicited from 8 native Xiangshan speakers (4 female, age range: 47-53, mean age: 50). All of them were born and raised in Xiangshan, and were reported to use Xiangshan as their major language over 75% of time in daily life. The speakers were asked to read a semi-randomized order of test tokens as naturally as possible. Each token was presented in a separate slide in a Microsoft PowerPoint presentation to avoid potential list effects. Recordings took place in a quiet room locally in Xiangshan, using an H4n Zoom recorder at a sampling rate of 44.1k Hz. To prevent possible code-switching or influence from other languages, all the instructions were given solely in Xiangshan by the first author, who is also a native Xiangshan speaker.

To look specifically at the variations in lexical tone sandhi, this paper selected 112 f_0 contours of 14 disyllabic lexical compounds produced by all 8 speakers. There are 9 distinctive syllables in the initial position of the disyllables, which come from 2 historical tonal categories (*yinping* and *yinshang*), and were all pronounced as an HH tone by the informant in the pilot study. The second syllables with the same synchronic tone almost all come from the same historical tonal category (i.e., HH from *yinping*, HL from *yinqu*, LHL from *yangping* except for 链 ‘chain’ from *yangqu*, and LH from *yangqu*). The disyllables were designed to cover all the 4 possible non-checked citation tone combinations with an initial HH tone (i.e., HH-HH, HH-HL, HH-LHL, HH-LH), as shown in Table 2. We excluded 6 tokens due to unnatural production or background noise. In total, 106 f_0 contours were examined in this paper.

Table 2 Selected disyllabic lexical compounds in this study. The first column indicates the synchronic and historical tonal category of the initial syllable. The first row indicates those of the second syllable. The synchronic tones values were produced by the informant in the pilot study.

	HH <i>yinping</i>	HL <i>yinqu</i>	LHL <i>yangping</i>	LH <i>yangqu</i>
HH	书包	青菜	书房	青豆
<i>yin</i>	<i>backpack</i>	<i>a variety</i>	<i>study (N.)</i>	<i>green</i>
<i>ping</i>	樱花 <i>cherry blossom</i> 青椒 <i>green pepper</i>	of <i>Chinese cabbage</i>	樱桃 <i>cherry</i>	soya <i>bean</i>
HH	手机	海带	手链	扁豆
<i>yin</i>	<i>mobile</i>	<i>kelp</i>	<i>bracelet</i>	<i>haricot</i>
<i>shang</i>	phone 苦瓜 <i>bitter</i> gourd		水牛 <i>buffalo</i> 草莓 <i>strawberry</i>	<i>bean</i>

2.2. Data analysis

The recordings were segmented manually in *Praat* [18]. For each elicited token, the boundaries of the pitch-bearing unit of each syllable, which include onset glides, vowels, and final nasals, were marked. F_0 values at 10 equidistant points of each rhyme were obtained using *ProsodyPro* [19]. To normalize between-speaker variations, z-scores of logarithmic transformed f_0 were calculated based on the given speaker’s mean f_0 value [20], which were then used for further analysis.

Independent human perceptual categorisation and *k*-means clustering analysis were adopted to classify f_0 sandhi contours into distinctive clusters. Machine clustering, which is based on interpretable similarity metrics, is able to provide relatively objective and reproducible results. However, the outcome machine clusters may not always be perceptually relevant, as this method solely relies on the distance variance within the acoustic data. We combined the two approaches to cross-validate one another (i.e., producing converging classification results).

The perceptual analysis was conducted by the first author, who carefully classified the f_0 contours according to the perceptual and visual similarities. The *k*-means analysis for longitudinal data was performed using the package *kml* [21] in *R* [22]. It partitions the f_0 contours into the nearest clusters based on Euclidean distance (L2 norm) from the cluster centroids, which were refined through iterative optimization. The *k*-means clustering was run through 2 to 10 clusters with an initialization method of *kmeans*- followed by an alternation of *kmeans--* and *randomK*. It should be noted that *k*-means clustering works best when the number of clusters is predefined. Although the *kml* package provides non-parametric criteria for selecting the optimal number of clusters, which select the solution that maximizes the ratio of between-cluster variance to within-cluster variance [23], it is acknowledged by the package developers that ‘the determination of the optimal number of clusters is still an unsettled issue’ [23]. In this paper, we will present and compare the *k*-means cluster results with the machine-evaluated optimal number as obtained from the *kml* package, as well as those with the human-evaluated optimal number that comes from the perceptual analysis.

Based on the clustering results, we then examined several potential factors that could contribute to the variations of f_0 tone sandhi contours: historical tonal categories of the initial tone, synchronic citation tonal categories of the initial tone, synchronic citation tonal categories of the non-initial tone, speaker, and item. Other factors, such as morphosyntax, lexical frequency, and length, etc., were not considered in the current analysis, because the selection of materials was strictly controlled to include only disyllabic lexical compounds which are commonly occurring and frequently used. Due to the small sample size, only descriptive analyses were conducted at this stage, but we are planning to implement statistical analyses on the larger dataset in the future.

3. Results

3.1. Monosyllabic citation tones

The 9 distinctive monosyllables in the initial position of the selected disyllabic lexical compounds, although all pronounced with an HH tone by the informant in the pilot study, show inconsistent tone values among the 8 speakers (also discussed in Section 1). Three out of the 8 speakers (S4, S5, S8) pronounced some tokens with an HL tone, as shown in Table 3.

The tokens differ among the speakers, indicating a random rather than item-based nature of the citation tonal variations. In total, there are 18 tokens with an initial HL synchronic citation tone in the current dataset.

Table 3 Syllables pronounced with an HL tone

Speakers	Tokens produced with HL tone
S4	扁 <i>flat</i> , 水 <i>water</i> , 青 <i>green</i>
S5	书 <i>book</i> , 苦 <i>bitter</i> , 青 <i>green</i> , 樱 <i>cherry</i>
S8	扁 <i>flat</i> , 手 <i>hand</i> , 青 <i>green</i>

3.2. Disyllabic lexical tone sandhi patterns

As discussed in Section 2.2, both perceptual and k -means analyses were adopted to explore the categorical sandhi patterns in the selected disyllabic lexical compounds. The perceptual analysis revealed 4 distinctive sandhi f_0 contours in the data (Figure 1A). We identify the 4 clusters here with the heuristic phonetic, pre-phonological labels HMML, MMMH, MHHL, and HHHH, with the first and the last pairs of letters respectively describing the contours of the first and the second sandhi tones.

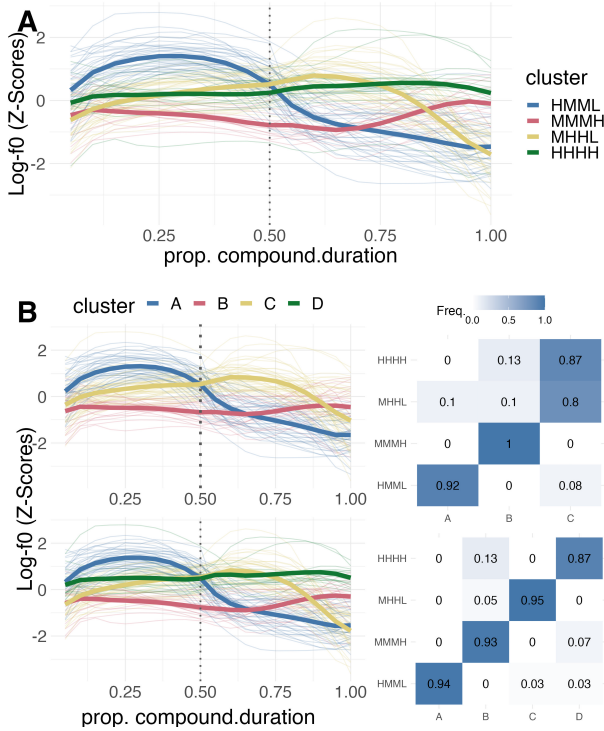


Figure 1 Cluster results of the perceptual analysis (A) and k -means clustering (B). Solid lines are cluster means and faint lines are individual f_0 contours. Dotted vertical lines indicate syllable boundaries. Heatmaps at right show the distribution of the k -means clusters (x-axis) against the perceptual clusters (y-axis).

As for the k -means clustering results, the optimal clustering according to Ray Turi criteria [24] in the *kml* package gives 3 clusters (Figure 1B, row 1), which are similar to the HMML, MMMH, and MHHL clusters in the perceptual analysis. The “missing” HHHH cluster was mostly (87%) partitioned into the cluster C, a rise-fall contour. The close distance between the

HHHH and MHHL clusters, whose first 75% portion of the contours almost overlap with each other, could be the reason why they are treated as similar contours by the machine which relies heavily on the distance matrix.

We then ran the k -means clustering with a predefined number of clusters of 4, the number obtained from the perceptual analysis. The outcome results are very similar to those identified in the perceptual analysis, with 4 similarly-shaped sandhi contours (Figure 1B, row 2). Furthermore, the distribution shows a neat one-to-one correspondence between the machine and human perceptual clusters, with over 87% overlap for each cluster pair (i.e., cluster A and HMML, cluster B and MMMH, cluster C and MHHL, cluster D and HHHH). The k -means clustering and human perceptual analysis led to mutually compatible results when the number of cluster is predefined for the k -means clustering, which mutually validate the accuracy of both methodological approaches. The automatization of choosing the optimal number using *kml* package can be problematic in this study, as it completely leaves out one perceptually distinctive category. To this end, we will use the results of the 4 clusters for further analysis. The mismatched cases between the two clustering methods were further examined. It should be noted that certain degrees of ambiguity are unavoidable, as such purely form-driven categorization essentially forces gradient f_0 contours into distinct categories. In this case, we believe the machine clustering can deal better with those “marginal” data points in each category via rigorous calculation of distance. Therefore, we largely rely on the 4 cluster results given by k -means analysis and only changed the cluster label when the investigated f_0 contour is perceptually very different from the category to which it had been assigned.

3.3. Possible factors in tone sandhi variation

We first examined the distribution of all four f_0 clusters across the synchronic citation tonal categories of the non-initial tones. By doing so, we aimed to explore whether those f_0 sandhi trajectories align with the anticipated left-dominant pattern in lexical compounds as attested in other Northern Wu dialects. That is, if such scenario is also relevant in Xiangshan, the citation tones of the non-initial tones should not play a role in varying the f_0 sandhi categories of the whole disyllables.

All the 4 f_0 sandhi clusters are found in each tonal category of the non-initial syllable (Figure 2A). Although the frequencies vary slightly across the 4 tonal categories, it is obvious that there is no one-to-one correspondence between the tonal categories and the f_0 clusters, suggesting that the citation tones of the non-initial tones cannot be the major source of such variations in sandhi outputs. The historical status of these tonal categories was not considered, as they are largely equal to the synchronic categories (see Section 2.1). To understand whether the various f_0 sandhi clusters can be explained by the initial tones, we then looked at the distribution of clusters across different historical and synchronic tonal categories of the initial tones. As shown in Figure 2B, an asymmetry of sandhi patterns is found between the *yinping* and *yinshang* tonal categories. While tokens with *yinping* initial tones mostly belong to clusters A and B (97.5% for HH and 91.7% for HL), those with *yinshang* initial tones have clusters C and D as their only sandhi clusters. The distinctiveness and exclusiveness in distributions imply a significant contribution of historical tonal categories to the variations of f_0 sandhi clusters. Such trends do not vary too much across synchronic citation tonal categories of the initial tones, suggesting that the synchronic tonal categories do not

serve as a major source of such sandhi variations. The only noticeable differences exist in the second most predominant sandhi clusters: cluster B for *yinping* only occurs twice for HL initial tone, and cluster D for *yinshang* is not available for HL initial tone at all. However, the relatively small number of HL tones in this dataset does not allow us to conclusively determine whether such difference in distributions of the 2 sandhi outputs is truly relevant.

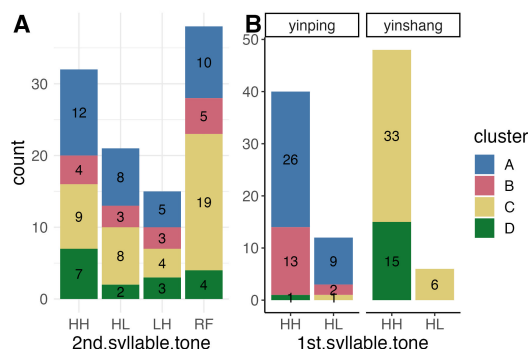


Figure 2 Distribution of clusters across different citation tones of the second syllable (A) and the first syllable (B)

Finally, the distributions of f_0 sandhi clusters across speakers, as well as items, were analysed. As shown in Figure 3, inter-speaker variations appear to be largely accountable for the presence of 2 distinct sandhi contours associated with each historical tonal category. Five out of the 8 speakers mostly produced cluster A contour for tokens with *yinping* initial tones, 4 of which (S1, S5, S6, S7) only produced this contour. The remaining 3 speakers produced cluster B contour as the most frequent pattern instead. On the contrary, for tokens with *yinshang* initial tones, 3 speakers (S1, S5, S8) only produced cluster C contour, and S6 only produced cluster D contour. The remaining 4 speakers produced a mixture of clusters C and D.

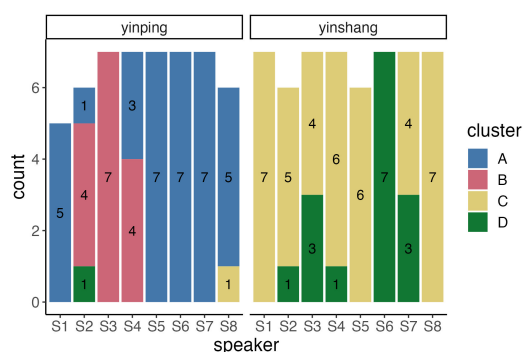


Figure 3 Distribution of clusters across speakers

Items, on the other hand, show much less influence on the variant sandhi patterns. No item was produced as one single cluster contour consistently across the speakers (Figure 4). Within the *yinping* category, clusters A and B are found as the first and second most predominant patterns for each token, which conforms to the overall trend found across all the tokens. The same is true for tokens under the *yinshang* category, but

with clusters C and D instead. Overall, this suggests that the selection of sandhi output is not item-based.

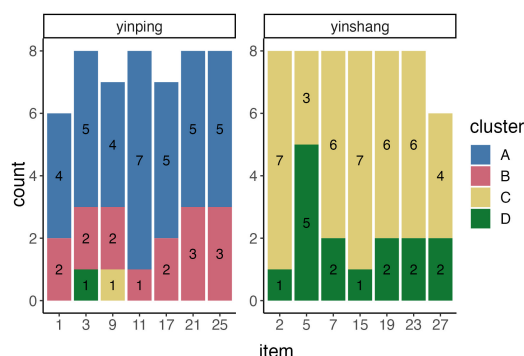


Figure 4 Distribution of clusters across items

4. Discussion

Our research discovered 4 distinctive sandhi contours in disyllabic lexical compounds with an initial H-register synchronic citation tone in Xiangshan Chinese: (a) HMML, a general fall; (b) MMMH, a general rise; (c) MHHL, a rise-fall; and (d) HHHH, two level tones. These findings were consistent across perceptual and k -means clustering analyses with 4 clusters.

The distributions of the 4 sandhi clusters suggest that the historical tonal categories of the initial tones, as well as individual speakers, contribute most to such categorical variations. In particular, disyllabic lexical compounds starting with a *yinping* historical tonal category mostly surface as an HMML or an MMMH contour, whereas those with an initial *yinshang* tonal category favour the other two sandhi outputs: MHHL and HHHH. Zooming into each historical tonal category, the choice between the two sandhi outputs is mostly speaker-specific. That is, some speakers prefer to produce an HMML for a *yinping*-initial disyllable, others choose the MMMH contour as the predominant pattern instead.

The results underline the importance of historical tonal categories of the initial tones in lexical sandhi applications in Xiangshan, and suggest distinctive development paths for the sandhi forms and the citation tones. The monosyllabic tone system in Xiangshan has merged *yinping* and *yinshang* categories into one synchronic HH tone, and tonal switches are found between the HH and HL tones. Despite the complex tonal development in monosyllables, the disyllabic lexical compounds consistently show divergent sandhi patterns that are correlated with the historical tonal categories of the initial tone. In this sense, the sandhi tones seem to be *preservatory* in the diachronic tone change path and provide evidence confirming the reality of the historical tonal categories.

Various interpretations can be made of the inter-speaker variations. One possibility is that the speakers can be viewed as being at different stages of language change. For example, in the 2 sandhi realisations of *yinping*-initial lexical disyllables, the HMML output can be derived in a left-to-right association from an initial HL synchronic tone, whereas the MMMH output is less transparent and might come from an earlier stage. If it is truly the case, speakers who chose to produce MMMH contour may be in an older stage. Minimally, the presence of different sandhi forms presents a challenge to Generative Phonology derivations of tone rules. We leave the actual derivation mechanisms for future research.

5. References

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